

Microlocal Morse Theory and Truncated Microsupport

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March 9, 2025*

Preface

In these notes, We study microlocal Morse theory, which is developed by Guillermou, Kashiwara, and Schapira in [GKS12], but replace microsupports with truncated microsupprts introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Truncated Microsupport

Definition 1.1. (i) Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if F satisfies the equivalent conditions in Prop.

(ii) aa

References

- [GKS12] Stéphane Guillermou, Masaki Kashiwara, Pierre Schapira, *Sheaf Quantization of Hamiltonian Isotopies and Applications to Nondisplaceability Problems*, Duke. Math. J. 161, No. 2, pp.201–245, 2012.
- [GP74] Victor Guillemin, Alan Pollack, *Differential Topology*, Prentice-Hall, 1974.
- [KMS03] Masaki Kashiwara, Teresa Monteiro Fernandes, Pierre Schapira, *Truncated Microsupprt and Holomorphic Solutions of D-modules*, Ann. Scient. Éc. Norm. Sup., série 4, tome 36, pp.583–399, 2003.
- [KS90] Masaki Kashiwara, Pierre Schapira, *Sheaves on Manifolds*, Grundlehren der Mathematischen Wissenschaften, 292, Springer, 1990.
- [Hat89] 服部晶夫, 多様体 増補版, 岩波全書 288, 岩波書店, 1989.
- [Sch85] Pierre Schapira, *Microdifferential Systems in the Complex Domain*, Grundlehren der Mathematischen Wissenschaften, 269, Springer, 1985.

*2025/03/09 first draft

- [ST92] Pierre Schapira, Nobuyuki Tose, *Morse Inequalities for $\mathbf{R}$ -constructible Sheaves*, Adv. Math. 93, pp.1–8, 1992.